Chapter 4: Coupled Microstrip Line

Outline

- Literature Examples
- Motivations
- Coupled Microstrip Line
- Coupled Microstrip Line Using Front-End Decoupling Capacitor
- Coupled Microstrip Line Using Distributed Decoupling Capacitors
- Conclusions

---

1. Literature Examples

(chap4.pdf, p.3–7)

Four prior art approaches to reduce far-end crosstalk (FEXT) on coupled microstrip lines are summarized below.

1.1 Guard Trace [Ref. 1]

(chap4.pdf, p.3)

A floating guard trace is placed between the aggressor (active line) and victim (quiet line) on the PCB surface.

- **Advantage:** Easy implementation with standard PCB process
- **Disadvantage:** Only a *slight* reduction of far-end crosstalk noise

  Figure description (p.3): 3-D view of three parallel PCB traces on FR4 substrate: aggressor (left), guard (center), victim (right). All sit on top of the dielectric. No vias present.

1.2 Grounded Guard Trace [Ref. 2]

(chap4.pdf, p.4)

A guard trace with via connections to ground is placed between aggressor and victim.

- **Advantage:** Efficient reduction of far-end crosstalk noise
- **Disadvantage:** Layout area is limited due to via placement constraints

  Figure description (p.4): Top-view schematic showing three horizontal traces (ports 1–4 at ends). The middle (guard) trace has periodic circular via symbols connecting it to ground.

1.3 Grounded Guard Trace with Dielectric Overlay [Ref. 3]

(chap4.pdf, p.5)

A grounded guard trace plus a dielectric superstrate (overlay) placed on top of the PCB structure.

- **Advantage:** Efficient FEXT reduction using only two vias
- **Disadvantage:** Additional cost of the dielectric overlay

Figure description (p.5): 3-D cross-section view showing Active Line, Guard Trace, and Victim Line on FR4 substrate with ε_r, overlaid with a dielectric layer of height h_l. Vias at each end of the guard trace connect to ground.

1.4 Serpentine Guard Trace [Ref. 4]

(chap4.pdf, p.6)

A serpentine-shaped guard trace (no vias, no dielectric overlay) placed between the two signal lines.

- **Advantage:** Efficient FEXT reduction without vias or dielectric overlay
- **Disadvantage:** Large PCB area required for the serpentine pattern

Figure description (p.6): 3-D view showing aggressor and victim parallel traces with a meandering serpentine trace between them, all on FR4.

1.5 Alternately Open-Circuited Stubs [Ref. 5]

(chap4.pdf, p.7)

Open-circuited stubs placed alternately along both the aggressor and victim lines.

- **Advantage:** Reduces the area compared to serpentine guard trace
- **Disadvantage:** Causes large reflections on the signal line

Figure description (p.7): Schematic showing two parallel transmission lines (Aggressor Line and Victim Line) each with alternating T-stub branches that are open-circuited. Source with V_{s1} and termination resistors R_0 at each end.

---

2. Motivations

(chap4.pdf, p.8)

The chapter proposes a new solution that addresses all the disadvantages of prior work simultaneously:

Goal	What to Eliminate
Save cost	Eliminate dielectric overlay
Increase routing flexibility	Eliminate vias
Save PCB area	Eliminate serpentine guard trace
Reduce reflections	Eliminate alternately open-circuited stubs

---

3. Coupled Microstrip Line (Baseline)

3.1 Topology and Parameters

(chap4.pdf, p.9)

- **Substrate:** FR4, $\varepsilon_r = 4.4$, $\tan\delta = 0.02$
- **Structure:** Active Line and Quiet Line as parallel microstrip traces

Dimensions:

w (mm)	s (mm)	h (mm)	l (mm)
1.42	0.7	0.8	40

where w = trace width, s = spacing between traces, h = substrate height, l = trace length.

Equivalent circuit per unit length: - Self inductance L_s per line - Mutual inductance L_m between lines - Self capacitance C_s per line (to ground) - Mutual capacitance C_m between lines

> **Figure description (p.9):** Left: 3-D view of two parallel orange traces on grey FR4, with labels "Active Line," "Quiet Line," "Near End," "Far End," rise time t_r, and length l. Center-right: cross-sectional view showing trace width w, spacing s, substrate height h, and ε_r. Far right: lumped equivalent circuit segment with series L_s and L_m (mutual) and shunt C_s and C_m.

3.2 Even- and Odd-Mode Equivalent Circuits

(chap4.pdf, p.10)

Even-mode equivalent circuit: Both lines driven in phase. Mutual inductor adds $(L_s + L_m)$, no current through C_m.

$$TD_{even} = l\sqrt{(L_s + L_m)(C_s)}$$

(chap4.pdf, p.10)

Odd-mode equivalent circuit: Lines driven out of phase. Mutual inductor subtracts $(L_s - L_m)$, capacitance doubles $(C_s + 2C_m)$.

$$TD_{odd} = l\sqrt{(L_s - L_m)(C_s + 2C_m)}$$

(chap4.pdf, p.10)

> **Figure description (p.10):** Left circuit (even): each line has series inductor $L_s + L_m$ and shunt capacitor C_s to ground; no coupling between lines. Right circuit (odd): each line has series inductor $L_s - L_m$ and shunt capacitors C_s (to ground) plus $2C_m$ (between lines). The two circuits are labeled "Active Line" and "Quiet Line."

3.3 FEXT and NEXT — Even/Odd Decomposition

(chap4.pdf, p.11)

The four-port system (Active: Port 1→Port 3; Quiet: Port 2→Port 4) is analyzed by superimposing even- and odd-mode excitations:

- **Direct excitation:** source V_s on active line only, Z_0 terminates all ports.
- **Even-mode excitation:** source $V_s/2$ on each line with same polarity; each line sees Z_{even}.
- **Odd-mode excitation:** source $+V_s/2$ on active and $-V_s/2$ on quiet line; each line sees Z_{odd}.

$$V_4(t) = V_{4e}(t) + V_{4o}(t)$$

(chap4.pdf, p.11)

$$V_2(t) = V_{2e}(t) + V_{2o}(t)$$

(chap4.pdf, p.11)

> **Figure description (p.11):** Three circuit diagrams stacked. Top (direct): single source V_s feeding Port 1 of Active Line through Z_0, Quiet Line grounded through Z_0 at Port 2; both far ends terminated in Z_0. Middle (even): sources $V_s/2$ on both lines, lines modeled as Z_{even}, ε_{even}. Bottom (odd): source $+V_s/2$ on active, $-V_s/2$ on quiet; lines modeled as Z_{odd}, ε_{odd}.

3.4 FEXT and NEXT — Ideal Coupled Line Results

(chap4.pdf, p.12–13)

Simulation results for the baseline structure:

TD_{even} (ps)	TD_{odd} (ps)	T_d (ps)
253.614	229.202	24.412

where $T_d = TD_{even} - TD_{odd}$ is the time difference between modes.

- **Far-end crosstalk noise (FEXT):** 0.076 V
- **Near-end crosstalk noise (NEXT):** 0.019 V

The FEXT waveform is a negative pulse of duration T_d arriving at $t = TD_{odd}$.

> **Figure description (p.12):** Left graph shows even-mode (positive ramp) and odd-mode (negative ramp) signals at far end vs. time; TD_{even} and TD_{odd} arrows show their respective arrival times, with T_d as the gap. Right graph shows the resulting FEXT $V_4(t)$: a negative spike of ~-0.076 V near $t = 0.27$ ns followed by a return to zero.

> **Figure description (p.13):** NEXT $V_2(t)$ plot: a trapezoidal positive pulse starting at $t = 0$, plateauing at ~0.019 V, lasting from 0 to ~0.5 ns ($= 2 \times TD$), then decaying.

3.5 FEXT and NEXT Estimation Equations

(chap4.pdf, p.14)

Peak of NEXT occurs at $t = t_r$:

$$V_2(t_r) = V_{2e}(t_r) + V_{2o}(t_r)$$

(chap4.pdf, p.14)

$$V_{2e}(t_r) = \frac{V_s}{2}\left(\frac{Z_{even}}{Z_{even} + Z_0}\right)$$

(chap4.pdf, p.14)

$$V_{2o}(t_r) = -\frac{V_s}{2}\left(\frac{Z_{odd}}{Z_{odd} + Z_0}\right)$$

(chap4.pdf, p.14)

Peak of FEXT occurs at $t = TD_{even}$:

$$V_4(t) = V_{4o}(t) + V_{4e}(t)$$

(chap4.pdf, p.14)

$$|V_4(TD_{even})| = V_{max} \cdot \min\left(\frac{T_d}{t_r},\, 1\right)$$

(chap4.pdf, p.14)

$$V_{max} = \frac{Z_0}{Z_{even} + 2Z_0 + Z_{odd}} V_s$$

(chap4.pdf, p.14)

> **Figure description (p.14):** Equations annotated on two boxed formulas. The NEXT box highlights $V_2(t_r) = V_{2e}(t_r) + V_{2o}(t_r)$. The FEXT box highlights $|V_4(TD_{even})| = V_{max} \min(T_d/t_r, 1)$.

3.6 Lossless Coupled Microstrip Line

(chap4.pdf, p.15–16)

When dispersion is included (lossless model with frequency-dependent ε_{eff}):

TD_{even} (ps)	TD_{odd} (ps)	T_d (ps)
254.414	234.002	20.412

- **FEXT:** 0.0641 V
- **NEXT:** 0.0169 V

> **Figure description (p.15):** Left graph: even and odd mode waveforms at far end, showing slight temporal shift due to dispersion (dashed "with dispersion" vs. solid "without dispersion"). Right graph: FEXT $V_4(t)$ showing close match between direct simulation and sum of even/odd components.

> **Figure description (p.16):** NEXT $V_2(t)$ for lossless case: plateau at ~0.0169 V from $t = 0$ to ~$2 \times TD \approx 0.5$ ns.

3.7 Lossy Coupled Microstrip Line

(chap4.pdf, p.17–18)

Including conductor and dielectric losses ($\tan\delta = 0.02$):

TD_{even} (ps)	TD_{odd} (ps)	T_d (ps)
258.814	238.202	20.612

	FEXT (V)	NEXT (V)
Simulation (ADS source)	0.063	0.0180
Simulation (measured source)	—	—
Measurement	0.055	0.0163

Figure description (p.17): Left: even/odd mode time waveforms showing amplitude reduction due to loss (dashed "With Loss" curves are smaller than solid "Without Loss"). Right: FEXT $V_4(t)$ comparing ADS simulation, simulation with measured source, and physical measurement — all closely match.

Figure description (p.18): NEXT $V_2(t)$ for lossy case with photo of real PCB circuit (two SMA connectors on each end of FR4 board with two parallel copper traces visible). Plot shows flat-top NEXT pulse agreeing between simulation and measurement.

3.8 Eye Diagram — Baseline

(chap4.pdf, p.19–21)

Simulation Setup: PRBS source V_s on active line; observe V_3 (far end of active line).

Eye diagram performance vs. bit rate:

Bit Rate	t_r	Eye High (%)	Eye Width (%)	Jitter (%)
2 Gbps	50 ps	97.2	99.6	4.44
4 Gbps	25 ps	95.2	99.6	4.44
8 Gbps	12.5 ps	94	99.76	4.432
16 Gbps	**6.25 ps**	**79.6**	**99.2**	**13.36**

At 16 Gbps the eye height degrades noticeably (79.6%) and jitter increases significantly (13.36%).

Figure description (p.20): Four eye diagram plots (voltage vs. time) for 2, 4, 8, and 16 Gbps. At lower rates the eye is wide open; at 16 Gbps the eye begins to close with visible jitter spreading.

---

4. Coupled Microstrip Line Using Front-End Decoupling Capacitor

4.1 Topology

(chap4.pdf, p.22)

Same FR4 substrate and trace dimensions as baseline ($\varepsilon_r = 4.4$, $\tan\delta = 0.02$, $w = 1.42$ mm, $s = 0.7$ mm, $h = 0.8$ mm, $l = 40$ mm), but a capacitor C_f is added between the two lines at the near end (source end).

In the equivalent circuit, the front-end capacitor appears as a distributed element C_f/l added in parallel with C_m across the entire line length.

> **Figure description (p.22):** Same 3-D and cross-sectional views as baseline, but the 3-D view shows a lumped capacitor C_f connected between the Active Line and Quiet Line at the Near End. The equivalent circuit segment now shows C_f/l in parallel with C_m.

4.2 Even- and Odd-Mode Equivalent Circuits with C_f

(chap4.pdf, p.23)

The even-mode circuit is **unchanged** (the capacitor C_f appears as a differential element and does not affect even mode):

$$TD_{even} = l\sqrt{(L_s + L_m)(C_s)}$$

(chap4.pdf, p.23)

The odd-mode circuit gains extra capacitance from $2C_f/l$ per unit length:

$$TD^c_{odd} = l\sqrt{\left(C_s + 2C_m + \frac{2C_f}{l}\right)(L_s - L_m)}$$

(chap4.pdf, p.23)

> **Figure description (p.23):** Left (even): unchanged from baseline — $L_s + L_m$ series inductor, C_s shunt capacitor, no C_f. Right (odd): $L_s - L_m$ series inductor, shunt capacitors $C_s + 2C_m + 2C_f/l$ in parallel.

4.3 Design Concept — Making $TD_{even} = TD^c_{odd}$

(chap4.pdf, p.24)

The key idea: choose C_f so that the modified odd-mode time delay equals the even-mode time delay, i.e., $T_d = 0$.

Define: - TD_{even}: even-mode delay (same with or without C_f) - TD_{odd}: odd-mode delay *without* C_f - TD^c_{odd}: odd-mode delay *with* C_f - T_d: original time difference $= TD_{even} - TD_{odd}$

Condition for zero FEXT:

$$T_d = TD_{even} - TD_{odd} = TD^c_{odd} - TD_{odd}$$

(chap4.pdf, p.24)

Solving for required C_f:

$$T_d = l\sqrt{L_s - L_m}\left(\sqrt{C_s + 2C_m + \frac{2C_f}{l}} - \sqrt{C_s + 2C_m}\right)$$

(chap4.pdf, p.24)

Rearranging to find the needed C_f that makes $TD^c_{odd} = TD_{even}$.

4.4 FEXT and NEXT — Even/Odd Decomposition with C_f

(chap4.pdf, p.25)

The decomposition is the same as before, but in the odd-mode circuit the front-end capacitor becomes $2C_f$ (one at each port of the odd-mode equivalent).

Design example: $T_d = 24.412$ ps $\rightarrow$ requires $C_f = 0.6$ pF.

> **Figure description (p.25):** Three circuit diagrams showing direct excitation (with C_f between ports), even-mode excitation (no C_f — unchanged), and odd-mode excitation (with $2C_f$ capacitors at both the source and far end of each line).

4.5 FEXT and NEXT — Ideal Coupled Line with C_f

(chap4.pdf, p.26–27)

With $C_f = 0.6$ pF, $T_d = 0$:

TD_{even} (ps)	TD^c_{odd} (ps)	T_d (ps)
253.614	253.614	**0**

- **FEXT:** reduced from 0.076 V $\rightarrow$ **0.027 V**
- **NEXT:** increased from 0.019 V $\rightarrow$ **0.080 V**

The FEXT is not zero even though $T_d = 0$ because of residual reflections from the lumped capacitor.

> **Figure description (p.26):** Left: even-mode signal (positive ramp) and odd-mode (negative step + exponential decay due to C_f RC effect) at far end — they arrive at the same time ($T_d = 0$) but have different shapes due to the capacitor. Right: FEXT $V_4(t)$ shows a much smaller disturbance (≈0.027 V peak) compared to baseline.
>
> **Figure description (p.27):** NEXT $V_2(t)$ with C_f shows a sharp initial spike to 0.080 V (due to the lumped capacitor charging current) followed by exponential decay.

4.6 FEXT and NEXT Estimation Equations with C_f

(chap4.pdf, p.28–29)

Odd-mode equivalent circuit at near end (Port 2):

The source $-V_s(t)/2$ charges capacitor $2C_f$ through the parallel combination of Z_0 and Z_{odd}. Defining:

$$V_{oc} = -\frac{V_s}{2t_r}\left(\frac{Z_{odd}}{Z_0 + Z_{odd}}\right), \qquad R_t = \frac{Z_0 Z_{odd}}{Z_0 + Z_{odd}}, \qquad \tau = 2R_t C_f$$

(chap4.pdf, p.28)

The near-end odd-mode voltage:

$$V_{2o}(t) = V_{oc}\tau\left[(e^{-t/\tau} - 1)u(t) - (e^{-(t-t_r)/\tau} - 1)u(t - t_r)\right] + V_{oc}(t)$$

(chap4.pdf, p.28)

Peak of NEXT at $t = t_r$:

$$V_2(t_r) = \frac{V_s}{2}\left(\frac{Z_{even}}{Z_{even} + Z_0}\right) + V_{oc}\tau(e^{-t_r/\tau} - 1) + V_{oc}t_r, \quad \tau = 2R_t C_f$$

(chap4.pdf, p.29)

Peak of FEXT at $t = TD_{odd} + T_d + t_r$:

$$V_4(t) = V_{4o}(t) + V_{4e}(t)$$

(chap4.pdf, p.29)

$$V_{4o}(t) = \left(\frac{2Z_0}{Z_0 + Z_{odd}}\right) V_{2o}(t - TD_{odd})$$

(chap4.pdf, p.29)

$$V_4(TD_{odd} + T_d + t_r) = \left(\frac{2Z_0}{Z_0 + Z_{odd}}\right)\left[V_{oc}\tau(e^{-(t_r+T_d)/\tau} - e^{-T_d/\tau}) + V_{oc}t_r\right] + V_{max}$$

(chap4.pdf, p.29)

4.7 FEXT and NEXT — Lossless Coupled Microstrip with C_f

(chap4.pdf, p.30–31)

With $T_d = 20.412$ ps (lossless case) $\rightarrow C_f = 0.5$ pF:

TD_{even} (ps)	TD^c_{odd} (ps)	T_d (ps)
254.414	254.414	**0**

- **FEXT:** reduced from 0.0641 V $\rightarrow$ **0.018 V**
- **NEXT:** 0.071 V (increased vs. baseline 0.0169 V)

4.8 FEXT and NEXT — Lossy Coupled Microstrip with C_f

(chap4.pdf, p.32–33)

With $T_d = 20.612$ ps (lossy) $\rightarrow C_f = 0.5$ pF:

TD_{even} (ps)	TD^c_{odd} (ps)	T_d (ps)
258.814	258.814	**0**

	FEXT (V)	NEXT (V)
Simulation	0.0257	0.0710
Measurement	0.0268	0.0636

> **Figure description (p.33):** Photo of real circuit with front-end C_f (slightly longer PCB board with surface-mount capacitor visible near one end). NEXT $V_2(t)$ shows the sharp initial spike characteristic of the charging C_f.

4.9 Eye Diagram with Front-End C_f

(chap4.pdf, p.34–36)

Bit Rate	t_r	Eye High (%)	Eye Width (%)	Jitter (%)
2 Gbps	50 ps	97.2	99.6	4.44
4 Gbps	25 ps	88.8	99.6	4.44
8 Gbps	12.5 ps	86	99.2	**8.88**
16 Gbps	**6.25 ps**	**64.4**	**97.44**	**31.2**

Comparison with baseline at 16 Gbps: Eye height worsens (64.4% vs. 79.6%), jitter worsens (31.2% vs. 13.36%). The front-end C_f reduces FEXT but *degrades* the eye diagram due to the NEXT spike and impedance mismatch.

4.10 S-Parameters with Front-End C_f

(chap4.pdf, p.37–38)

S-parameter comparison among: Coupled Microstrip Line (baseline), Front-End C_f, and Stub-Alternated [Ref. 5].

Reflection ($|S_{11}|$) and Near-End Coupling ($|S_{21}|$): - Front-End C_f shows much higher $|S_{11}|$ and $|S_{21}|$ (worse) than baseline or stub-alternated design across 0–6 GHz. - The baseline coupled microstrip line shows periodic notches (deep nulls) in both S_{11} and S_{21}.

Transmission ($|S_{31}|$) and Far-End Coupling ($|S_{41}|$): - Front-End C_f has slightly more insertion loss ($|S_{31}|$) than baseline due to the shunt capacitor. - $|S_{41}|$ (FEXT in frequency domain): Front-End C_f significantly improves (lowers) the far-end coupling compared to baseline.

> **Figure description (p.37):** Two frequency plots (0–6 GHz). Left: $|S_{11}|$ — baseline has oscillating notches (good), Front-End C_f is nearly flat near -10 dB (poor), Stub-Alternated has notches. Right: $|S_{21}|$ — similar pattern.

Figure description (p.38): Two frequency plots. Left: $|S_{31}|$ (transmission) — both lines similar, Front-End C_f slightly lower. Right: $|S_{41}|$ (FEXT) — Front-End C_f shows much lower coupling than baseline (-20 to -40 dB range vs. baseline -5 to -15 dB range).

---

5. Coupled Microstrip Line Using Distributed Decoupling Capacitors

5.1 Concept and Topology

(chap4.pdf, p.39)

Instead of one lumped C_f at the front end, the total capacitance $C_f = 0.5$ pF is distributed along the line as N smaller capacitors $\Delta C_f = C_f/N$ spaced at intervals Δl.

Design rule for spacing Δl: Use the guided wavelength at the 3-dB bandwidth frequency.

$$f_{3\text{dB}} = \frac{0.35}{t_r}$$

(chap4.pdf, p.39)

$$\lambda = \frac{3 \times 10^8}{f_{3\text{dB}}} \cdot \frac{1}{\sqrt{\varepsilon_{eff}}}$$

(chap4.pdf, p.39)

Distributed capacitor configurations (for $C_f = 0.5$ pF total):

Δl (mm)	Fraction of λ	Number of caps	ΔC_f (pF) each
9.2	$\lambda/2$	4	0.1250
4.6	$\lambda/4$	8	0.0625
1.9	$\lambda/10$	20	0.0250

Figure description (p.39): Circuit schematic showing Active Line and Quiet Line with multiple small capacitors ΔC_f placed at regular intervals Δl along the coupled region. Same Z_0 source and terminations as before.

5.2 FEXT and NEXT — Distributed C_f Results

(chap4.pdf, p.40)

Comparison of near- and far-end crosstalk for different Δl spacings vs. lumped C_f:

Configuration	FEXT (V)	NEXT (V)
Lumped C_f	0.0257	0.071
$\Delta l = \lambda/10$	**0**	**0.028**

The $\lambda/10$ distributed case achieves: - FEXT $\rightarrow$ effectively **0** (eliminated) - NEXT reduced from 0.071 V (lumped) to **0.028 V** (much improved)

Figure description (p.40): Left: NEXT $V_2(t)$ for lumped C_f, $\lambda/2$, $\lambda/4$, $\lambda/10$ spacings — as spacing decreases, the initial spike flattens and approaches the unmodified crosstalk shape but at much lower amplitude. Right: FEXT $V_4(t)$ — the $\lambda/10$ distributed case shows essentially zero far-end crosstalk.

5.3 S-Parameters — Distributed C_f

(chap4.pdf, p.41–42)

Reflection ($|S_{11}|$) and Near-End Coupling ($|S_{21}|$): - As spacing Δl decreases (more capacitors, finer distribution), $|S_{11}|$ improves (approaches baseline coupled microstrip performance). - The $\lambda/10$ distributed case nearly matches the baseline in reflection performance.

Transmission ($|S_{31}|$) and Far-End Coupling ($|S_{41}|$): - Distributed ΔC_f at $\lambda/10$ shows significantly better transmission (less insertion loss) than lumped C_f. - $|S_{41}|$ (far-end coupling) for $\lambda/10$ distributed case shows very deep notches (better FEXT suppression) across the 0–6 GHz band compared to all other configurations.

Figure description (p.41): S-parameter plots comparing 5 configurations: baseline, front-end C_f, $\lambda/2$, $\lambda/4$, $\lambda/10$ distributed, and Stub-Alternated [5]. The $\lambda/10$ distributed case tracks closely to the baseline for both $|S_{11}|$ and $|S_{21}|$.

Figure description (p.42): $|S_{31}|$ (transmission): distributed cases nearly identical to baseline. $|S_{41}|$ (far-end coupling): distributed $\lambda/10$ case achieves -50 to -80 dB suppression with periodic notches — significantly better than all prior art.

---

6. Conclusions

(chap4.pdf, p.43)

Final comparison table (lossy microstrip, $t_r = 40$ ps):

Configuration	FEXT (V) Meas.	FEXT (V) Sim.	NEXT (V) Meas.	NEXT (V) Sim.
Conventional	0.0550	0.0630	0.0163	0.0180
Front-end C_f	0.0268	0.0257	0.0636	0.0710
Distributed C_f	X (not meas.)	**0**	X (not meas.)	**0.0280**

Key conclusions:

1. **FEXT and NEXT:**
   - Front-end C_f reduces FEXT (by ~52%) but increases NEXT (by ~4×)
   - Distributed C_f ($\lambda/10$) reduces FEXT to **zero** and limits NEXT to 0.028 V
2. **Eye Diagram:**
   - Performance is *degraded* when a front-end (lumped) decoupling capacitor is used due to the large NEXT spike and impedance discontinuity.
   - Distributed capacitors restore eye performance closer to baseline.
3. **S-Parameters:**
   - Reflection and transmission performance are *restored* when using distributed (finer-spacing) capacitors.

- The $\lambda/10$ distributed case achieves far-end coupling ($|S_{41}|$) suppression to -50 dB or better across 0–6 GHz.

Key Equations Summary

Quantity	Formula	Reference
Even-mode TD (no C_f)	$TD_{even} = l\sqrt{(L_s + L_m)C_s}$	*(chap4.pdf, p.10)*
Odd-mode TD (no C_f)	$TD_{odd} = l\sqrt{(L_s - L_m)(C_s + 2C_m)}$	*(chap4.pdf, p.10)*
Odd-mode TD (with C_f)	$TD^c_{odd} = l\sqrt{(C_s + 2C_m + 2C_f/l)(L_s - L_m)}$	*(chap4.pdf, p.23)*
NEXT peak	$V_2(t_r) = \frac{V_s}{2}\frac{Z_{even}}{Z_{even}+Z_0} - \frac{V_s}{2}\frac{Z_{odd}}{Z_{odd}+Z_0}$	*(chap4.pdf, p.14)*
FEXT peak magnitude	$\lVert V_4(TD_{even})\rVert = V_{max}\min(T_d/t_r, 1)$	*(chap4.pdf, p.14)*
V_{max} for FEXT	$V_{max} = \frac{Z_0}{Z_{even}+2Z_0+Z_{odd}}V_s$	*(chap4.pdf, p.14)*
Required T_d compensation	$T_d = l\sqrt{L_s - L_m}\left(\sqrt{C_s + 2C_m + \frac{2C_f}{l}} - \sqrt{C_s + 2C_m}\right)$	*(chap4.pdf, p.24)*
3-dB frequency	$f_{3\text{dB}} = 0.35/t_r$	*(chap4.pdf, p.39)*
Guided wavelength	$\lambda = \frac{3\times10^8}{f_{3\text{dB}}\sqrt{\varepsilon_{eff}}}$	*(chap4.pdf, p.39)*
RC time constant (C_f circuit)	$\tau = 2R_tC_f$, where $R_t = Z_0Z_{odd}/(Z_0 + Z_{odd})$	*(chap4.pdf, p.28)*